

3.2 The Integers

Investment in The Kraft Heinz Company (Fair Value: March 31, 2017 - \$29.55; December 31, 2016 - \$28.43)	29,550	28,430
Receivables	10,000	10,000
Inventories	10,000	10,000
Property, plant and equipment	10,000	10,000
Goodwill	10,000	10,000
Other intangible assets	10,000	10,000
Deferred charges reinsurance assumed	10,000	10,000
Other	10,000	10,000
Railroad, Utilities and Energy:		
Cash and cash equivalents	1,000	1,000
Property, plant and equipment	10,000	10,000
Goodwill	10,000	10,000
Regulatory assets	10,000	10,000
Other	10,000	10,000
Finance and Financial Products:		
Cash and cash equivalents and U.S. Treasury Bills:		
Cash and cash equivalents	10,000	10,000
U.S. Treasury Bills	10,000	10,000
Total cash, cash equivalents and U.S. Treasury Bills	20,000	20,000
Investments in equity and fixed maturity securities	10,000	10,000
Other investments	10,000	10,000

Figure 3.11 A ledger comparing assets to debts, resulting in net wealth. (credit: modification of work “Reviewing Financial Statements” by Mary Cullen/Flickr, CC BY 2.0)

After completing this section, you should be able to:

1. Define and identify numbers that are integers.
2. Graph integers on a number line.
3. Compare integers.
4. Compute the absolute value of an integer.
5. Add and subtract integers.
6. Multiply and divide integers.

Positive net wealth is when the total value of a person's assets, such as their home, their 401(k), their car, and savings account balance, exceed that of their debts, such as car loans, mortgages, or credit card debt. However, when the total value of debt exceeds the total value of assets, then the person has negative net wealth. Expressing the negative net wealth as a negative number allows people to work with the positive net values and negative net values with the same mathematical processes, and in the same applications. This section introduces the integers and operations with integers.

Defining and Identifying Integers

Extending the counting numbers to include negative numbers and zero forms the **integers**. Any other number that cannot be written as $\{\dots - 3, -2, -1, 0, 1, 2, 3, \dots\}$ is not an integer.

EXAMPLE 3.26

Identifying Integers

Which of the following are integers and which are not?

-3	Is an integer, as it is the negative of a counting number
$\sqrt{24}$	This is not written as an integer. Entering the square root of 24 in a calculator, such as desmos, the result is 4.899 (rounded off). Since this is not an integer, then $\sqrt{24}$ is not an integer.
$36/4$	Since 36 divided by 4 is 9, and 9 is an integer, then $36/4$ is an integer.
45	Is an integer, as it is a counting number
63.9	Is not an integer, because it is not a counting number and not the negative of a counting number.

$2/7$	Dividing 2 by 7 results in a number less than 1, but greater than 0, so is between two consecutive integers. So, $2/7$ is not an integer.
-16.0	Is an integer, since the decimal part is 0

> YOUR TURN 3.26

Which of the following are integers?

1. -214
2. $38/11$
3. $\sqrt{90}$
4. $\sqrt{121}$
5. $420/35$

Graphing Integers on a Number Line

Integers are often imagined as steps along a path. You start at 0, and going to the left is going backward, or in the negative direction, while going to the right is going forward, or in the positive direction. A number line (Figure 3.12) helps envision the integers. This also means that an integer gives magnitude (size) and direction (positive is to the right, negative is to the left). Graphing an integer on the number line means placing a solid dot at the integer on the number line.

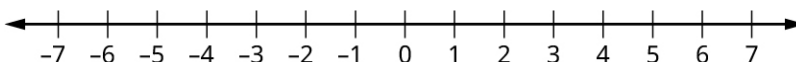


Figure 3.12

▶ VIDEO

[Graphing Integers on the Number Line \(https://openstax.org/r/Graphing_Integers\)](https://openstax.org/r/Graphing_Integers)

EXAMPLE 3.27

Graphing Numbers on the Number Line

Graph the following on the number line:

1. 1
2. -4
3. 3

✓ Solution

1.

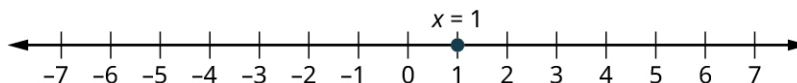


Figure 3.13

2.

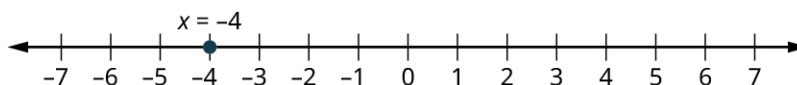


Figure 3.14

3.

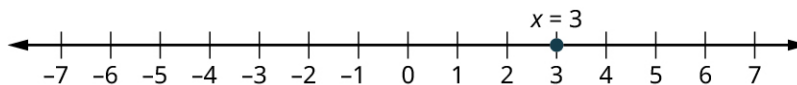


Figure 3.15

YOUR TURN 3.27

Graph the following numbers on the number line:

1. -10
2. 4
3. 0

Comparing Integers

When determining if one quantity or size is larger than another, we know it means there is more of whatever is being discussed. In terms of positive integers, we can envision that larger integers are further to the right on the number line. This idea applies to negative integers also. This means that **a is greater than b** when a is to the right of b on the number line. We write $a > b$. When a is greater than b , we can also say that **b is less than a** . On the number line, b would be to the left of a . We write $b < a$.

We need to recognize that $a > b$ means the same thing as $b < a$. This can be seen on the number line in [Figure 3.16](#). On this number line, a is to the right of b , so $a > b$. But this means b is to the left of a , so $b < a$.



Figure 3.16

VIDEO

[Comparing Integers Using the Number Line \(https://openstax.org/r/the_Number_Line\)](https://openstax.org/r/the_Number_Line)

EXAMPLE 3.28

Comparing Integers Using a Number Line

Determine which of -6 and 4 is larger using a number line, and express that using both the greater than and the less than notations.

Solution

To illustrate this, we use a number line ([Figure 3.17](#)).

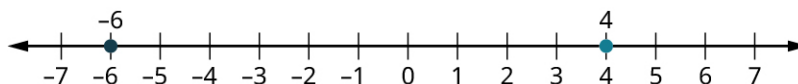


Figure 3.17

Since -6 is to the left of 4 , then -6 is less than 4 . We can write this as $-6 < 4$. Another way of expressing this is that 4 is greater than -6 . So we can also write $4 > -6$.

YOUR TURN 3.28

1. Determine which of -38 and 27 is larger using a number line, and express that using both the greater than and the less than notations.

EXAMPLE 3.29

Comparing Negative Integers

Determine which of -6 and -2 is larger, and express that using both the greater than and the less than notations.

Solution

To illustrate this, we use a number line ([Figure 3.18](#)).

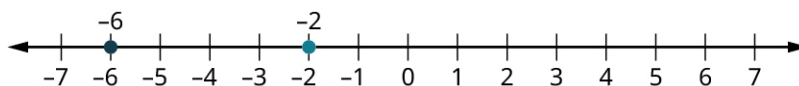


Figure 3.18

Since -6 is to the left of -2 , then -6 is less than -2 . We can write this as $-6 < -2$.

Another way of expressing this is that -2 is greater than -6 . So we can also write $-2 > -6$.

Warning: People often ignore the negative signs, and think that since 6 is greater than 2, -6 is greater than -2 . To avoid that error, remember that the greater number is to the right on the number line.

YOUR TURN 3.29

1. Determine which of -63 and -213 is larger, and express that using both the greater than and the less than notations.

EXAMPLE 3.30

Comparing Integers by Quantity

Determine which of 27 and 410 is larger, and express that using both the greater than and the less than notations.

Solution

When thinking about quantity, 410 is more than 27. So, 410 is greater than 27 and 27 is less than 410. We can write this as $410 > 27$ or as $27 < 410$.

YOUR TURN 3.30

1. Determine which of 101 and 98 is larger, and express that using both the greater than and the less than notations.

The Absolute Value of an Integer

When talking about graphing integers on the number line, one interpretation suggests it is like walking along a path. Negative is going to the left of 0, and positive going to the right. If you take 30 steps to the right, you are 30 steps away from 0. On the other hand, when you take 30 steps to the left, you are still 30 steps away from 0. So, in a way, even though one is negative and the other positive, these two numbers, 30 and -30 , are equal since both are 30 steps away from 0. The **absolute value of an integer** n is the distance from n to 0, regardless of the direction. The notation for absolute value of the integer n is $|n|$.

If we think of an integer as both direction and magnitude (size), absolute value is the magnitude part.

Calculating the absolute value of an integer is very straightforward. If the integer is positive, then the absolute value of the integer is just the integer itself. If the integer is negative, then to compute the absolute value of the integer, simply remove the negative sign. Keeping in mind the number line as a path, when you've gone 10 steps to the left of 0, you have still taken 10 steps, and the direction does not matter.

VIDEO

Evaluating the Absolute Value of an Integer (https://openstax.org/r/Absolute_Value_of_an_Integer)

EXAMPLE 3.31**Calculating the Absolute Value of a Positive Integer**

Calculate $|19|$.

✓ **Solution**

Since the number inside the absolute value symbol is positive, the absolute value is just the number itself. So $|19| = 19$.

> YOUR TURN 3.31

1. Calculate $|38|$.

EXAMPLE 3.32**Calculating the Absolute Value of a Negative Integer**

Calculate $|-435|$.

✓ **Solution**

Since the number inside the absolute value is negative, the absolute value removes the negative sign. So $|-435| = 435$.

> YOUR TURN 3.32

1. Calculate $|-81|$.

Adding and Subtracting Integers

You may recall having approached adding and subtracting integers using the number line from earlier in your academic life. Adding a positive integer results in moving to the right on the number line. Adding a negative integer results in moving to the left. Subtracting a positive integer results in a move to the left on the number line. But subtracting a negative integer results in a move to the right.

This leads to a few adding and subtracting rules, such as:

Rule 1: Subtracting a negative is the same as adding a positive.

Rule 2: Adding two negative integers always results in a negative integer.

Rule 3: Adding two positive integers always results in a positive integer.

Rule 4: The sign when adding integers with opposite signs is the same as the integer with the larger absolute value.

These rules are good to keep in the back of your mind, as they can serve as a quick error check when you use a calculator.

EXAMPLE 3.33**Adding Integers**

Use your calculator to calculate $4 + (-7)$. Explain how the answer agrees with what was expected.

✓ **Solution**

Using a calculator, we find that $4 + (-7) = -3$. Since we are adding integers with opposite signs, the sign of the answer matches the sign of the integer with the larger absolute value which $|-7| = 7$.

> YOUR TURN 3.33

1. Use your calculator to calculate $(-18) + 11$. Explain how the answer agrees with what was expected.

EXAMPLE 3.34**Subtracting Positive Integers**

Use your calculator to calculate $18 - 9$. Explain how the answer agrees with what was expected.

✓ Solution

Using a calculator, we find that $18 - 9 = 9$. Since 18 was larger than 9, we expected the difference to be positive.

> YOUR TURN 3.34

1. Use your calculator to calculate $38 - 100$. Explain how the answer agrees with what was expected.

EXAMPLE 3.35**Subtracting with Negative Integers**

Use your calculator to calculate $27 - (-13)$. Explain how the answer agrees with what was expected.

✓ Solution

Using a calculator, we find that $27 - (-13) = 40$. Since we're subtracting a negative number, it is the same as adding a positive, so this is the same as $27 + 13 = 40$.

> YOUR TURN 3.35

1. Use your calculator to calculate $45 - (-26)$. Explain how the answer agrees with what was expected.

EXAMPLE 3.36**Adding Integers with Opposite Signs**

Use your calculator to calculate $(-13) + 90$. Explain how the answer agrees with what was expected.

✓ Solution

Using a calculator, we find that $(-13) + 90 = 77$. Since we are adding integers with opposite signs, the sign of the answer matches the sign of the integer with the larger absolute value, which is positive since 90 is positive.

> YOUR TURN 3.36

1. Use your calculator to calculate $19 + (-36)$. Explain how the answer agrees with what was expected.

One use of negative numbers is determining **net worth**, which is all the wealth someone owns less all that someone owes. Sometimes net worth is positive (which is good), and sometimes net worth is negative (which can be stressful).

EXAMPLE 3.37**Calculating Net Worth**

Jennifer is owed \$50 from her friend Janice, but owes her friend Pat \$87. What is Jennifer's net worth?

 Solution

Net worth is the amount that one is owed minus the amount one owes. Jennifer is owed \$50 but owes \$87. So, her net worth is $50 - 87 = -37$. The negative indicates that Jennifer owes more than she is owed.

 YOUR TURN 3.37

1. Christian is owed \$180 from his friend Chanel, but owes his friend Jeff \$91. What is Christian's net worth?

Multiplying and Dividing Integers

Similar to addition and subtraction, the signs of the integers impact the results when multiplying and dividing integers. The rules are fairly straightforward, but again rely on the direction on the number line. There are only two rules.

Rule 1: When multiplying or dividing two integers with the same sign, the result is positive.

Rule 2: When multiplying or dividing two integers with opposite signs, the result is negative.

Just as before, these rules can serve as a quick error check when using a calculator.

EXAMPLE 3.38**Multiplying Positive Integers**

Use your calculator to calculate 4×8 . Explain how the answer agrees with what was expected.

 Solution

Entering 4×8 into your calculator, the result is 32. This agrees with our expectation. The numbers have the same signs, so the result is positive.

 YOUR TURN 3.38

1. Use your calculator to calculate 81×26 . Explain how the answer agrees with what was expected.

EXAMPLE 3.39**Multiplying Integers with Different Signs**

Use your calculator to calculate $9 \times (-10)$. Explain how the answer agrees with what was expected.

 Solution

Entering $9 \times (-10)$ into your calculator, the result is -90 . This agrees with our expectation. The numbers have opposite signs, so the result is negative.

 YOUR TURN 3.39

1. Use your calculator to calculate $(-18) \times 13$. Explain how the answer agrees with what was expected.

EXAMPLE 3.40**Dividing Integers with Different Signs**

Use your calculator to calculate $400/(-25)$. Explain how the answer agrees with what was expected.

 Solution

Entering $400/(-25)$ into your calculator, the result is -16 . This agrees with our expectation. The numbers have opposite

signs, so the result is negative.

YOUR TURN 3.40

1. Use your calculator to calculate $(-116)/4$. Explain how the answer agrees with what was expected.

EXAMPLE 3.41

Dividing Negative Integers

Use your calculator to calculate $-750/(-3)$. Explain how the answer agrees with what was expected.

Solution

Entering $-750/(-3)$ into your calculator, the result is 250. This agrees with our expectation. The numbers have the same signs, so the result is positive.

YOUR TURN 3.41

1. Use your calculator to calculate $(-77)/(-11)$. Explain how the answer agrees with what was expected.

At the end of a season, a team may wish to buy their coach an end-of season gift. It makes sense to share the cost equally among the members. To do so, the team would need to find the average (or mean) cost per member. The **average (or mean) of a set of** numbers is the sum of the numbers divided by the number values that are being averaged.

EXAMPLE 3.42

Finding the Average of a Set of Numbers

The daily low temperatures in Barrie, Ontario, for the week of February 14, 2021, were -20° , -12° , -15° , -23° , -17° , -13° , and -19° degrees Celsius. What was the average daily temperature for the week of February 14, 2021, in Barrie?

Solution

Step 1: To find the average daily temperature, we first need to add the temperatures.

$$(-20) + (-12) + (-15) + (-23) + (-17) + (-13) + (-19) = -119$$

Step 2: That sum will then be divided by 7 since we are averaging over seven days, giving $-119/7 = -17$. So, the average daily temperature in Barrie, Ontario the week of February 14, 2021, was -17° Celsius.

YOUR TURN 3.42

1. Banks and credit cards often base their interest on the average daily balance of an account, which is the average of the balance from each day of the period. The account balance of Jada's checking account on each day of the week of December 13, 2020, was \$1,250, \$673, $-\$1,500$, \$1,000, \$785, \$785, and \$710. What was Jada's average daily balance for the week of December 13, 2020? Assume Jada pays no fees for a negative balance.

Check Your Understanding

6. Identify which of the following numbers are integers:
 -4 , 15.2 , $\sqrt{2}$, $\frac{3}{20}$, 430
7. Graph the following integers on the number line: 4 , -2 , 7 .
8. Place these integers in increasing order: 4 , -2 , -7 , 10 , -13 .

9. Calculate $|-7|$.
10. Calculate $4 - (-9)$.
11. Calculate $(-3) \times (-12)$.



SECTION 3.2 EXERCISES

1. Identify all the integers in the following list: 4, -17, 8, 0.5, $\sqrt{7}$, $\frac{1}{9}$, -300.
2. Identify all integers in the following list: -9.2, 13, -1, $\sqrt{47}$, $\frac{\sqrt{3}}{2}$, 567, -300.

For the following exercises, plot the integers on the same number line.

3. 4, -2, 10, 0
4. -6, -3, 10, 1, 4
5. -3, -10, 7, 2
6. 2, 4, 8, -2, -5

For the following exercises, determine if the comparison is true or false.

7. $-3 < -10$
8. $-10 > -3$
9. $7 > -6$
10. $-6 < 7$
11. $18 < 20$
12. $20 < 18$
13. What are two numbers with an absolute value of 76?
14. What are two numbers with an absolute value 87?
15. Determine $|-67|$.
16. Determine $|98|$.
17. Determine $|61|$.
18. Determine $|-903|$.
19. What are two numbers that are 23 away from 0?
20. What are two numbers 13 away from 0?

For the following exercises, complete the indicated calculation.

21. $47 + 200$
22. $67 + (-86)$
23. $(-86) + 104$
24. $13 - (-54)$
25. $(-45) - (-26)$
26. $(-13) + (-102)$
27. 38×12
28. $(-12) \times (-10)$
29. $14 \times (-5)$
30. $421 \times (-13)$
31. $(-412) \times 504$
32. $(-8,919) \times (-3,401)$
33. $(-130) \div (-65)$
34. $450 \div 9$
35. $2000 \div (-40)$
36. $(-910) \div 14$
37. The daily low temperatures, in degrees centigrade, in Fargo for the week of January 17, 2021, were -9, -17, -18, -14, -17, -19, and -11. What was the average low temperature in Fargo that week?
38. Riley collects checks for a fundraiser supporting the homeless in town. Through Venmo, they collect the following amounts: \$20, \$20, \$50, \$75, \$250, \$10, \$15, \$65, \$30, \$15. What was the average donation that Riley collected?
39. Heath has \$495 in an account. They will collect two paychecks this week, one for \$150 and the other for \$250. Heath also will pay three bills, one for \$50, one for \$110, and one for \$300. After all those transactions, how much will Heath have in their account?
40. Five diners decide to split the check evenly. The total bill comes to \$475. How much does each diner owe?